Evaluating FedAvg and FedProx for Federated Crop Disease Detection under Data Heterogeneity

A Simulated Collaborative Agricultural Network with Non-IID Foliar Distributions

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Real-World Motivation: Agricultural Impact & Data Silos

The Agronomic and Global Food Security Context:

- Crop foliar diseases account for **20% to 40% annual yield reduction** across staple crops worldwide (Savary et al., *Nat. Ecol. Evol.*, 2019).
- Early computer vision diagnosis can **cut chemical pesticide misuse by up to 50%** (FAO, 2022).

Why Centralized Machine Learning Fails in Agricultural Reality:

Commercial Confidentiality & Sovereignty

Photographic records of crop health, infestations, and yield are sensitive trade secrets; producers resist cloud exposure.

Rural WAN Bandwidth Bottlenecks

Farms operate over constrained cellular/satellite uplinks; transmitting gigabytes of high-res foliar photography is unfeasible.

The Distributed Paradigm: Federated Learning (FL) enables decentralized collaborative training where raw foliar images remain on-premises and only numerical model parameters circulate.

The Distributed Challenge: Non-IID Foliar Skew & Client Drift

Agricultural Data is Intrinsically Non-IID (Statistical Heterogeneity):

Farm A (Humid Valley)

Dominant: Necrotic Blights (Rust, Early/Late Blight).
Negligible bacterial spots.

Farm B (Highland Zone)

Dominant: Bacterial Pathologies (Bacterial Spot, Mold).
Negligible fungal blights.

Farm C (Lowland Fields)

Dominant: Viral Infections & Healthy Foliage.
Negligible necrotic blights.

Sensor and Covariate Shift: Ambient solar illumination, soil backgrounds, and smartphone optics.

The Distributed Systems Consequence: Client Drift

Local gradient updates steer parameters toward local optima that conflict with the global objective. Standard **FedAvg** exhibits severe round-to-round oscillation, slow convergence, and **catastrophic accuracy degradation on minority disease classes**.

Research Question & Formal Hypotheses

*"To what extent does **FedProx** improve diagnostic performance, consensus stability, and cross-client generalization compared to **FedAvg** in a distributed plant disease detection system when three farm nodes possess statistically heterogeneous (non-IID) data distributions?"*

Hypothesis 1 (Performance Equity)

Under IID data, FedAvg and FedProx achieve parity; under acute non-IID skew, FedProx achieves superior macro-F1 and preserves minority class recall.

Hypothesis 2 (Optimization Stability)

FedProx dampens inter-round oscillations via its proximal anchor, yielding lower inter-round variance $S_{\Delta acc}$.

Hypothesis 3 (Parameter Sensitivity)

An intermediate proximal penalty ($\mu \in [0.01, 0.1]$) balances consensus stability vs. optimization stalling ($\mu \geq 1.0$).

Hypothesis 4 (Fault Tolerance under Dropout)

When a node disconnects (round $t = 5$), FedProx retains higher parameter stability and diagnostic resilience.

Distributed Foundations: Protocols & Optimization

Distributed Coordination Principles:

- **Star Topology & RPC**: Coordinator orchestrates workers via gRPC messages (`FitIns`, `FitRes`).
- **Barrier Latency Model**:

$$T_{\text{round}}^{(t)} = \max_{k \in \mathcal{K}_t} \{ T_{\text{down}}^{(k)} + T_{\text{comp}}^{(k)} + T_{\text{up}}^{(k)} \} + T_{\text{agg}}$$

- **Communication Complexity**: Exchanging serialized weights ($B_{\text{round}} = 8Kd$ bytes) reduces network payload by $> 98\%$ vs. raw image transmission.

Mathematical Optimization Formulation:

- **Baseline: FedAvg (McMahan et al., 2017)**

$$w^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{(t+1)}$$

- **Investigated: FedProx (Li et al., 2020)**

$$\min_w F_k(w) + \frac{\mu}{2} \left\| w - w^{(t)} \right\|_2^2$$

Elastic proximal penalty pulls divergent worker gradients toward global centroid.

Understanding the Barrier Latency Formula

Barrier Latency Model — Round Completion Time:

$$T_{\text{round}}^{(t)} = \max_{k \in \mathcal{K}_t} \left\{ T_{\text{down}}^{(k)} + T_{\text{comp}}^{(k)} + T_{\text{up}}^{(k)} \right\} + T_{\text{agg}}$$

Each symbol represents the following:

Symbol	Meaning
$T_{\text{round}}^{(t)}$	Total time of round t (from broadcast to aggregation completion).
$\mathcal{K}_t$	Set of participating clients in that round.
$T_{\text{down}}^{(k)}$	Time for client k to **receive** (download) the global model.
$T_{\text{comp}}^{(k)}$	Time for client k to **train locally** (E SGD epochs).
$T_{\text{up}}^{(k)}$	Time for client k to **upload** its updated parameters to the server.
T_{agg}	Time the server needs to **aggregate** (combine) all client updates.

Farm Nodes as Distributed Compute Nodes

Shared-Nothing Architecture: Each farm is an autonomous node with private storage, independent compute, and an isolated model replica. No shared memory or filesystem — communication exclusively via serialized message passing.

Node Properties

- **Local State Isolation**: Node k maintains private state $\sigma_k = (w_k, \mathcal{D}_k, n_k, t_k)$. No other node can read or modify it.
- **Independent Failure Domain**: Crash at Farm A does not propagate to B or C — *failure isolation* across fault domains.
- **Stateless Interactions**: Each `FitIns`/`FitRes` is self-contained — enables transparent node replacement.

Scalability & Consensus

- **Horizontal Scalability**: Adding node $K{+}1$ requires only $\mathcal{O}(Kd)$ additional aggregation cost — no data repartitioning.
- **Relaxed State Replication**: Unlike Paxos/Raft, federated consensus permits optimistic local execution + post-hoc reconciliation.
- **Aggregation as Consensus**: $w^{(t+1)} = \sum_k \frac{n_k}{n} w_k^{(t+1)}$ merges divergent replicas into global state.

Node Lifecycle & Distributed Fault Scenarios

Node State Machine (per Farm):

1. **IDLE** — Awaiting `FitIns` RPC.
2. **SYNCHRONIZING** — Deserializing $w^{(t)}$; overwriting local replica.
3. **TRAINING** — Executing E local SGD epochs; state *diverges* from global.
4. **REPORTING** — Serializing $w_k^{(t+1)}$ into `FitRes`; uploading to coordinator.
5. **PARTITIONED** — Network loss or crash detected via timeout τ_{timeout}; node excluded from quorum $\mathcal{S}_t$.

Recovery: Partitioned node re-joins by receiving next round's $w^{(t+1)}$ — *state catch-up* via parameter overwrite.

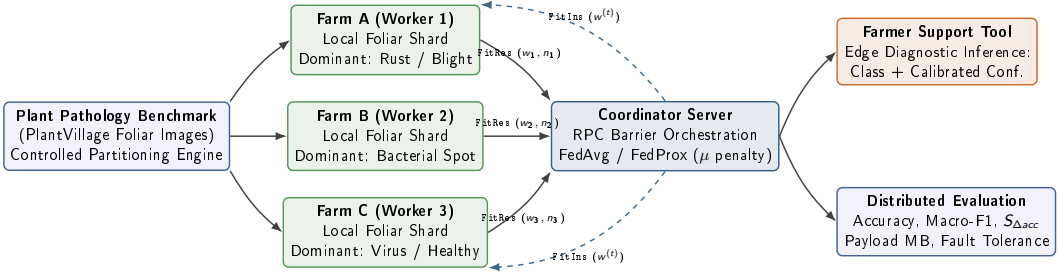
Fault Tolerance Mechanisms:

- **Network Partitions:** Timeout τ_{timeout} distinguishes straggler vs. unreachable node.
- **Quorum-Based Aggregation:**

$$w^{(t+1)} = \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j} w_k^{(t+1)}$$

- **Majority Quorum:** $|\mathcal{S}_t| \geq \lceil K/2 \rceil + 1$ prevents split-brain divergence.
- **Byzantine Awareness:** Crash-stop model assumed; Krum/trimmed-mean aggregation supports $f < K/3$ malicious nodes (future work).

Collaborative Farm Network Architecture



Distributed Decoupling: Private photographs never leave local worker boundaries; only serialized parameter tensors and sample counts cross the network channels.

Controlled Non-IID Partitioning & Edge Support Tool

Four Rigorous Partitioning Regimes:

1. **IID Baseline ($\alpha \to \infty$):** Balanced uniform distribution ($n_k = N/3, p_k \approx p_0$).
2. **Low Non-IID ($\alpha = 1.0$):** Dirichlet distribution with mild sample variations.
3. **Medium Non-IID ($\alpha = 0.5$):** Dirichlet distribution with moderate class concentration ($< 10\%$ in minority classes).
4. **High Non-IID (85/15 Disjoint Rule):** Extreme geographic skew where each farm receives 85% of an exclusive disease cluster and 7.5% from others.

Farmer Support Tool & Calibration:

- **Edge Input:** Technician captures foliar photo in field.
- **Model Inference:** Local feedforward pass on compiled global model.
- **Confidence Calibration:** Standard softmax is overconfident (Guo et al., 2017); evaluated via post-hoc **Temperature Scaling** and **Expected Calibration Error (ECE)**.
- **Agronomic Advisory:** Specific treatment guidance (fungicide, isolation, monitoring).

Targeted Experimental Matrix (6 Systematic Tracks)

1. FedAvg vs. FedProx Optimization

Compares global diagnostic accuracy, loss trajectories, and convergence speed across rounds.

2. Statistical Heterogeneity Impact

Quantifies accuracy penalty across $\alpha \in \{\infty, 1.0, 0.5, \text{disjoint}\}$ as cross-farm skew intensifies.

3. Proximal Parameter Sweep

Evaluates $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ to detect optimal balance vs. local update stalling.

4. Fault Tolerance & Node Churn

Evaluates system recovery and consensus degradation when Farm C disconnects at round $t = 5$.

5. Communication Complexity

Profiles serialized payload (MB/round), barrier completion latency, and rural WAN viability.

6. Cross-Node & Minority Equity

Assesses per-client test performance and class-wise recall to guarantee equity for small farms.

Stability Metric: $S_{\Delta acc} = \sqrt{\frac{1}{R-1}\sum_{r=2}^{R}(\Delta acc_r - \overline{\Delta acc})^2}$ (lower indicates smoother optimization).

Preliminary Results: RabbitMQ Prototype & Quorum Validation

Empirical Verification of Distributed Coordinator-Worker Prototype:

Scenario	Quorum	Samples	Global $w^{(1)}$	Barrier Latency (T)	Queue State & Behavior
Complete Round	3/3	N=60	**2.8333**	**120 ms** (T_{agg}=12.5 ms)	Queues drained; manual acks confirmed.
Node Dropout	2/3	N=30	**1.6667**	**3.00 s** (τ timeout barrier)	Dynamic re-weighting; C task queued.

Nominal Aggregation ($K = 3$)

$$w^{(1)} = \frac{10(1) + 20(2) + 30(4)}{60} = \frac{170}{60} \approx 2.8333$$

Barrier Latency: $T_{\text{barrier}} \approx 120$ ms; $T_{\text{agg}} = 12.5$ ms. Unblocks immediately upon complete quorum.

Dropout Resilience ($K = 2$, Farm C down)

$$w^{(1)}_{\text{dropout}} = \frac{10(1) + 20(2)}{30} = \frac{50}{30} \approx 1.6667$$

Barrier Latency: $T_{\text{barrier}} = 3.00$ s (fault detection via τ). $T_{\text{agg}} = 8.2$ ms; coordinator advances without deadlock.

Distributed Systems Theory in Federated Agriculture

CAP Theorem Trade-offs (Brewer, 2000):

- **Consistency:** All nodes see identical global model.
- **Availability:** Every farm can train locally, even offline.
- **Partition Tolerance:** System survives network failures.

Design Choice: AP System

Our federated network prioritizes **Availability + Partition Tolerance**. Farms train autonomously; coordinator aggregates when reachable. **Eventual consistency** is achieved over successive communication rounds.

Message-Passing & Logical Clocks:

- **RabbitMQ AMQP 0-9-1:** Durable direct/topic exchanges, persistent queues, manual acks (`FitIns`, `FitRes`).
- **Causal Ordering ($\prec$):**

$$\text{broadcast} \prec \text{train}_k \prec \text{upload} \prec \text{aggregate}$$

- **Duplicate Suppression:** Monotonic round IDs prevent duplicate uploads from biasing aggregation.
- **Idempotent Aggregation:** Deterministic weighted reduction ensures reproducibility.

Deliverables & Academic Architecture Strategy

Software Pipeline & Modular Separation:

- **Distributed Training Simulation:** Implemented in **Python**, **PyTorch**, and **RabbitMQ (AMQP 0-9-1)** for multi-process worker execution, state serialization, and non-IID partitioning.
- **Edge Diagnostic Engine (High-Efficiency C++ Deployment):**
  - Trained global CNN exported via **ONNX** / **TorchScript**.
  - Standalone inference runtime using **ONNX Runtime C++ API** / LibTorch.
  - Zero Python dependencies on edge device: deploys natively on Raspberry Pi or agricultural field gateways with sub-50ms inference latency.

Project Deliverables:

- Reproducible GitHub repository: `https://github.com/darioXxx32/IA`.
- Automated scripts for partitioning, multi-client simulation, evaluation logs, and visual analytics.

Project Feasibility, Work Schedule & Reproducibility

14-Week Milestone Timeline (Phases 1–6):

Phase	Milestone & Core Activities	Weeks	Status
Phase 1	AMQP RabbitMQ Middleware, Quorum & Unit Tests	W1–3	Completed
Phase 2	PlantVillage Dirichlet Partitioning ($\alpha \in \{\infty, 1.0, 0.5, \text{disjoint}\}$)	W4–5	Completed
Phase 3	CNN Architecture & PyTorch FedAvg/FedProx Loops	W6–7	In Progress
Phase 4	Systematic Benchmark of 6 Tracks (μ sweep, node dropout)	W8–10	Planned
Phase 5	ONNX Edge Compilation & Calibrated Advisory CLI	W11–12	Planned
Phase 6	Comparative Analysis & Camera-Ready Scientific Report	W13–14	Planned

Open-Science Reproducibility & Codebase

Codebase Repository: `https://github.com/darioXxx32/IA`
Protocol: Deterministic seeds (`seed=42`) and automated unit test suite (`test_protocol.py`).

Roadmap for Final Report & Expected Contributions

Roadmap for the Final Report:

1. Complete empirical execution of the 6 experimental tracks across all 4 data skew regimes.
2. Systematic parameter sweep over $\mu \in \{0.001, 0.01, 0.1, 1.0\}$, recording macro-F1, per-class recall, and stability metric $S_{\Delta acc}$.
3. Stress-test fail-stop worker disconnection at round $t = 5$ to measure distributed fault tolerance.
4. Standalone ONNX compilation and interactive farmer decision-support CLI tool.
5. Final 8–15 page scientific report following required MDPI journal formatting (*Agronomy / Applied Sciences*).

Core Expected Contributions

Bridges distributed systems engineering (barrier synchronization, fault tolerance, communication complexity) and federated machine learning under agricultural data heterogeneity, guaranteeing diagnostic equity for smallholder farms.

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Thank you!

Questions & Discussion

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